“On the way with Python”

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Lecture

07

Loop in Python

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Introduction

- A loop is used to execute a block of code repeatedly as long as a condition is satisfied or for a fixed number of times. Loops help reduce code repetition and make programs efficient.
- In Python, loops are mainly of two types:

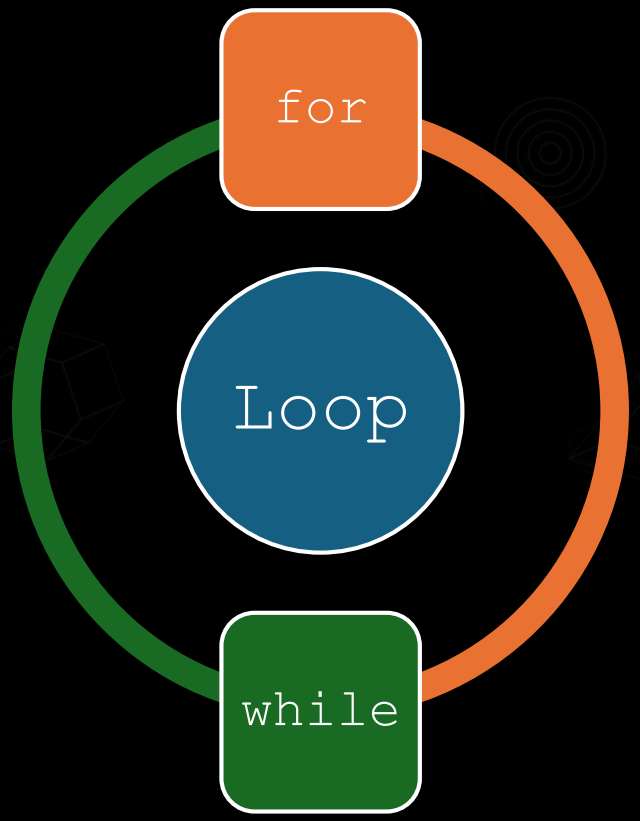
Why use Loop?

Loops are used to:

- Repeat tasks automatically
- Process multiple items in a sequence
- Reduce code duplication
- Perform calculations on large data sets

Types of Loops

- In Python, loops are mainly of two types:



Python for Loop

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for Loop

- With the for loop we can execute a set of statements. for loop used to iterate over a sequence (list, tuple, string, range, etc.).

for-loop.py

```
for variable in sequence:
    # code block
```

for Loop with Python range()

In Python, the **range()** function returns a sequence of numbers.

for-loop.py

```python
#Example: 1
for x in range(4):
  print(x)
#Example: 2
for x in range(2, 4):
  print(x)
```

output.txt

```
#Output: 1
0
1
2
3

#Output: 2
2
3
```

Loop Through a String

- Iterate over each character in language

for-loop.py

```python
language = 'Python'

# iterate over each character in language
for x in language:
    print(x)
```

output.txt

```
P
y
t
h
o
n
```

Loop Through a List

Access element of the list one by one.

for-loop.py

```python
languages = ['Swift', 'Python', 'Go']

# access elements of the list one by one
for lang in languages:
    print(lang)
```

output.txt

```
Swift
Python
Go
```

Loop Through a Dictionary

a) Loop Through Keys

for-loop.py

```python
my_dict = {
  "name": "Rohim",
  "age": 25,
  "city": "Dhaka"
}
for key in my_dict:
    print(value)
```

output.txt

```
name
age
city
```

Loop Through a Dictionary

b) Loop Through Values

for-loop.py

```python
my_dict = {
  "name": "Rohim",
  "age": 25,
  "city": "Dhaka"
}
for value in my_dict.values():
    print(value)
```

output.txt

```
Rohim
25
Dhaka
```

Loop Through a Dictionary

c) Loop Through Key-Value Pair

for-loop.py

```python
my_dict = {
  "name": "Rohim",
  "age": 25,
  "city": "Dhaka"
}
for key, value in my_dict.items():
    print(key, ":", value)
```

output.txt

```
name : Shakil
age : 25
city : Dhaka
```

Break Statement

The **break** statement terminates the **for loop** immediately before it loops through all the items.

for-loop.py

```python
languages = ['Swift', 'Python', 'Go', 'C++']

for lang in languages:
    if lang == 'Go':
        break
    print(lang)
```

output.txt

```
Swift
Python
```

Continue Statement

The **continue** statement skips the current iteration of the loop and continues with the next iteration.

for-loop.py

```python
languages = ['Swift', 'Python', 'Go', 'C++']

for lang in languages:
    if lang == 'Go':
        continue
    print(lang)
```

output.txt

```
Swift
Python
C++
```

For Loop Practice Task

1. Print **Odd** or **Even** from a list.
2. Print **Positive** or **Negative** from a list.
3. Print **Odd** or **Even** from a 1 to 20 must be used range.
4. **Sum** of the all numbers from a list.
5. **Count** how many **Odd** or **Even** numbers from 1 to 20
6. Find the **Largest Number** from a list.
7. Print the **Multiplication** table of number 5.
8. Sperate Odd and Even number form a list

"""Thanks All"""